Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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Outline

Motivation, question, headline

Setting. Europe moved electricity trading from 60-min to 15-min. Spain: two reforms in 2025.

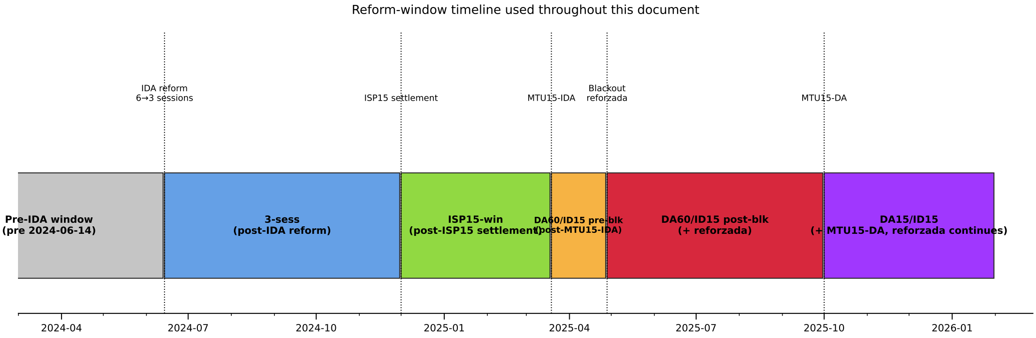
Question. Does finer granularity reshape *strategic* bidding? Who, when, which channel, market power up or down?

Headline. Granularity is a **strategic-latitude reveal**:

- Dispatchables widen ladders in critical hours.
- Granular-side prices fall; other market follows by anticipation.
- Wedge dispersion jumps and does not collapse.

$\Rightarrow$ Market power **compressed**, not amplified. Post-blackout cost rise is a separate story.

The Spanish reform sequence



- Three reforms in seven months: ISP15, ID15, DA15.
- April 2025 blackout → continuous *reforzada* regime.
- ID15 windows: pre-blackout. DA15 windows: hold reforzada constant by design.

Model sketch

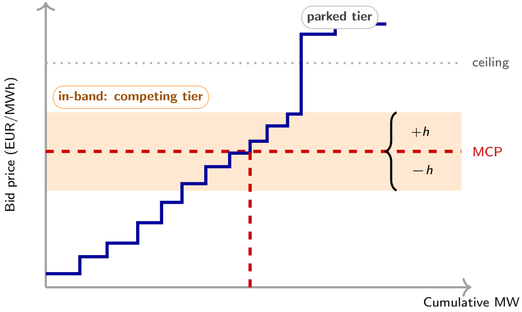
Setup. Sequential Cournot, extends Ito & Reguant (2016) with a granularity selector at each stage. Three agents: strategic bidder, operational aggregator, capacity-bound arbitrageur.

Reform path: $(1, 1) \to (1, Q) \to (Q, Q)$.

Key predictions to test:

- Bid ladders widen on the granular side.
- Aggregator shifts quantity into the granular market.
- Granular-side price drops; other market drops by anticipation.
- Cournot scale-up at (Q, Q).
- Wedge SD does *not* collapse at (Q, Q) — limited per-product arbitrage.
- Production-unit arbitrageurs exist and bind.

In-band region and bid-shape outcomes



In-band set (per curve u, d, p):

$$\mathcal{B} = \{k : |p_k - \mathrm{MCP}_{d,p}| \le h\}, \quad w_k = \frac{\Delta q_k}{\sum_{j\in\mathcal{B}} \Delta q_j}.$$

In-band weighted mean:

$$\bar{p} = \sum_{k\in\mathcal{B}} w_k\, p_k.$$

Bid-shape outcomes:

$$\sigma_p = \sqrt{\sum_{k\in\mathcal{B}} w_k\,(p_k - \bar{p})^2},$$

$$N_{\text{eff}} = \Big(\sum_{k\in\mathcal{B}} w_k^2\Big)^{-1}.$$

Bandwidth: $h = p_{90} - p_{50}$ of MCP per (window, market).

Identification

Data. OMIE bid-level (2022–2026), ENTSO-E weather, ESIOS ajuste costs.

Critical / flat partition from 15-min residual demand σ_{within}:

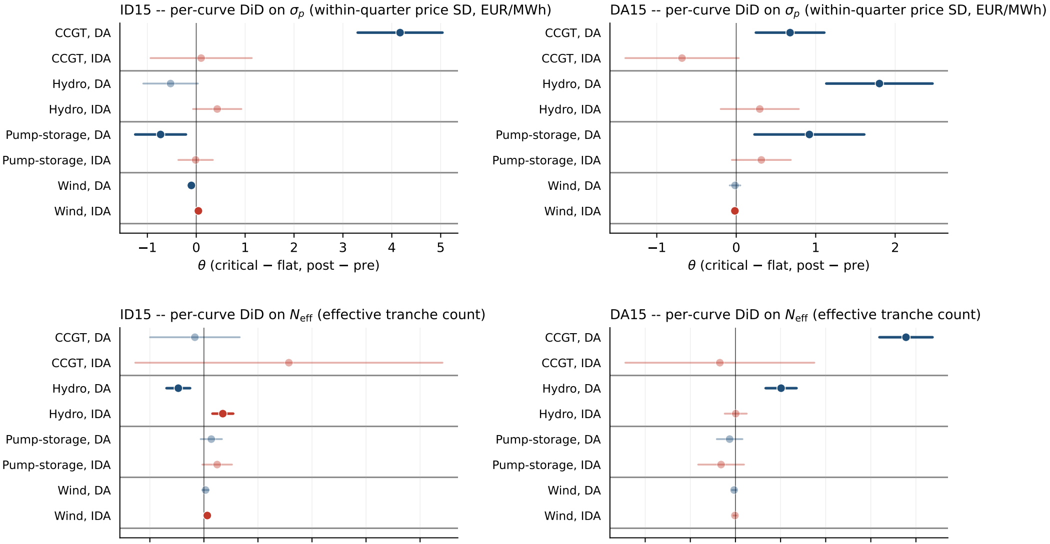
- Critical = morning + evening ramp.
- Flat = overnight $\Rightarrow$ within-day low-exposure control.

Three specifications:

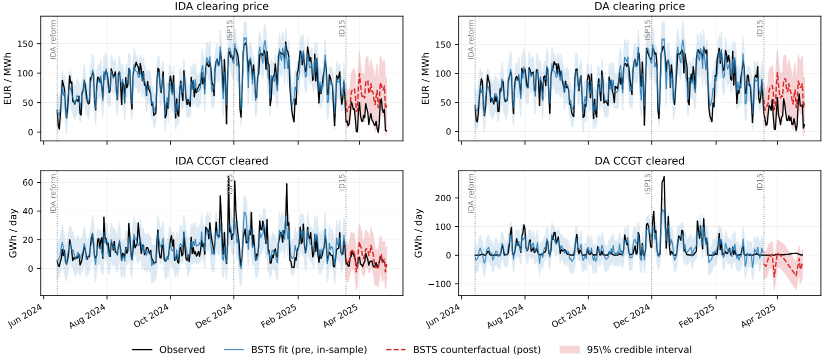
- Spec A — BSTS on prices, cleared MWh, wedge SD.
- Spec B — per-hour-class BSTS on in-band offered energy.
- Spec C — per-curve DiD on bid shape, critical $-$ flat differential.

Same-calendar placebos (2024, plus 2026 for ID15) net out recurring calendar shifts. DA15 pre-window starts at the blackout $\Rightarrow$ holds *reforzada* constant.

Result 1 — Bid ladders widen in critical hours

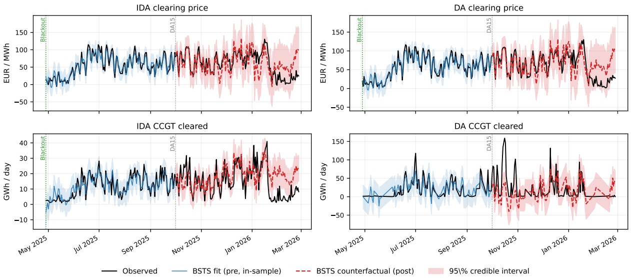


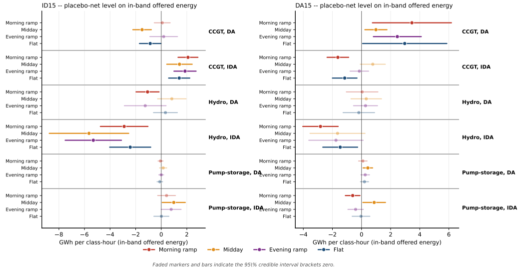
Result 2 — ID15 price drop in both markets



- Both DA and IDA prices drop on the same day, even though only IDA became finer.
- Cross-market symmetry = rational-anticipation channel.
- Spain-specific: France (same SIDC) does not move.

Result 3 — DA15 CCGT cleared scale-up (critical hours)





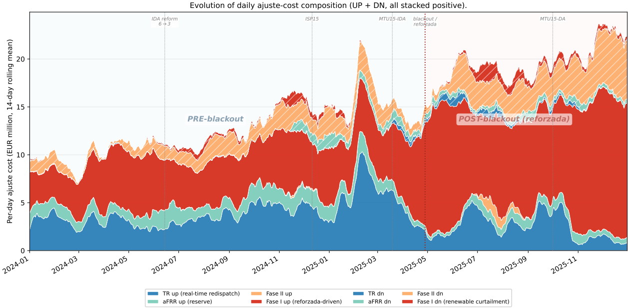
- Only level-shift surviving the joint placebo-net (+59 GWh/day).
- Concentrates in critical hours (right panel).
- DA only; IDA arm unchanged ⇒ migration into DA.

Result 4 — Wedge SD jumps, does not collapse

Wedge SD placebo-net (EUR/MWh)	ID15	DA15
Overall	$+2.16^{**}$	null
Critical hours	$+3.14^{***}$	null (non-reversion)

- Wedge *mean*: null. Wedge *SD*: jumps under ID15.
- DA15 does not revert it $\Rightarrow$ consistent with limited per-product arbitrage ($\sigma_\mu/(4b_1)$), not the strategic $\sigma_\mu/(8b_1)$).
- Bonus: ISP15 alone produces a same-sign jump $\Rightarrow$ bidders respond to the settlement reference itself.

Conclusion — granularity is a strategic-latitude reveal



Granularity channels shrink. Reforzada Fase I inflates.

Market power: *compressed*.

Consumers: auction channel favourable.

Granularity $\neq$ reforzada. The cost rise is post-blackout, CCGT-only — not MTU15.

Takeaway: granularity reveals *who* discriminates, *where*, and *when* within the day.

Thank you. Questions?

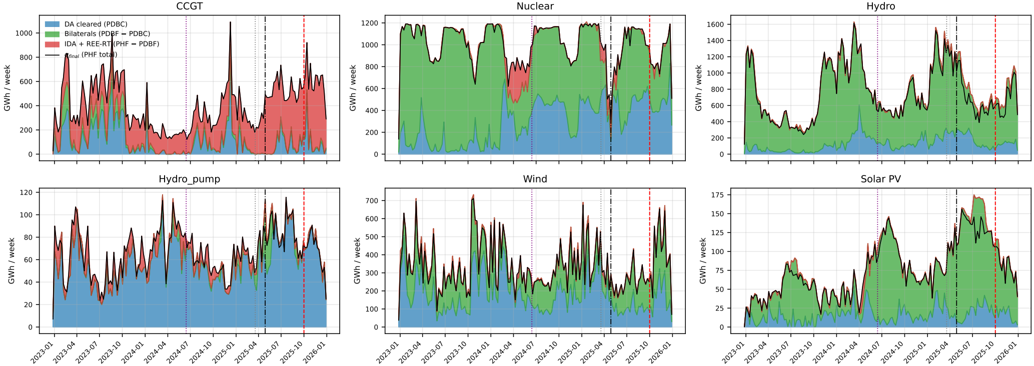
Backup — Production-unit arbitrageurs (P7)

Tech-group	n unit-sessions	% dual	Mean sell MW	Mean buy MW	Buy/Sell
Solar PV	2 093 392	26.9	54	55	**101%**
Wind	2 313 441	16.2	133	167	**126%**
Cogen	334 852	3.8	85	92	108%
Biomass	112 548	6.4	27	28	104%
Hydro	339 500	5.7	8 654	3 711	43%
Hydro pump	53 763	4.6	3 462	1 737	50%
CCGT	266 085	4.0	4 625	2 001	43%
Nuclear	13 054	4.1	1 225	360	29%

- VRE / small: buy $\approx$ sell $\Rightarrow$ forecast-driven rebalancing.
- Thermal / storage: buy/sell $\in [23\%, 50\%]$ $\Rightarrow$ sell-dominated rebalancing.
- Arbitrageur active across every tech.

Backup — Reforzada cost is CCGT-only (post-blackout)

Three-component decomposition (WEEKLY) per technology: DA + Bilaterals + IDA+REE-RT = q_{final}.



- CCGT day-ahead-cleared layer (blue) collapses post-blackout; IDA + REE-RT (red) balloons.
- Other technologies flat across the same window.
- $\Rightarrow$ CCGT-specific intervention, not a uniform reform effect.

Backup — Robustness summary

- **Pre-trend placebo.** ID15 IDA σ_p clean. DA15 DA: CCGT clean; pump-storage not separable from trend.
- **Cross-border + reforzada.** Hourly cross-border control leaves per-tech DiDs essentially unchanged. Reforzada acts on Fase I level, not within-hour SD.
- **Per-IDA-session BSTS.** ID15 price drop near-identical across S1, S2, S3.
- **Per-firm.** DA15 CCGT σ_p widening driven by Naturgy and Iberdrola.
- **Offer-type.** Restricting to simple offers, CCGT σ_p DiD rises to 3.53^{***}.